Formulario · los 9 pasos del LGDR

Todas despejadas · Laboratorio LGDR · Nise, secciones 8.5–8.6

PLANTA Y ECUACIÓN CARACTERÍSTICA

$$G(s) = \frac{K\,N(s)}{D(s)}$$

$$D(s) + K\,N(s) = 0$$

$$K = -\frac{D(s)}{N(s)}$$

$$N(s) = -\frac{D(s)}{K}$$

$$T(s) = \frac{K\,N(s)}{D(s) + K\,N(s)}$$

$$\sum p_{lc} = -a_{n-1}$$

$$\Rightarrow\ p_3 = -a_2 - 2\,\mathrm{Re}(s_0)$$

$$\prod(-p_{lc}) = a_0$$

PASO 1 · NÚMERO DE RAMAS

$$\text{ramas} = n$$

$$(n \geq m)$$

$$n = \text{ramas}$$

$$m = n - (\text{ramas a } \infty)$$

PASO 2 · SIMETRÍA

$$s = a + jb \in \text{LGDR}$$

$$\Rightarrow\ s^* = a - jb \in \text{LGDR}$$

PASO 3 · TRAMOS DEL EJE REAL

$$\sigma \in \text{LGDR}$$

$$\iff N_{der}(\sigma) = \text{impar}$$

$$N_{der}(\sigma) = \#\{\, p_i,\ z_i \text{ reales} > \sigma \,\}$$

$$\sum \angle(\text{par conjugado}) = 0^\circ$$

PASO 4 · INICIO Y TÉRMINO

$$K = 0$$

$$\Rightarrow\ D(s) = 0$$

$$(\text{polos})$$

$$K \to \infty$$

$$\Rightarrow\ N(s) = 0$$

$$(\text{ceros})$$

$$\text{ramas a } \infty = n - m$$

$$m = n - (\text{ramas a } \infty)$$

$$n = m + (\text{ramas a } \infty)$$

PASO 5 · ASÍNTOTAS

$$\sigma_a = \frac{\sum \mathrm{Re}\, p_i - \sum \mathrm{Re}\, z_i}{n - m}$$

$$\sum \mathrm{Re}\, p_i = \sigma_a(n-m) + \sum \mathrm{Re}\, z_i$$

$$\sum \mathrm{Re}\, z_i = \sum \mathrm{Re}\, p_i - \sigma_a(n-m)$$

$$n - m = \frac{\sum \mathrm{Re}\, p_i - \sum \mathrm{Re}\, z_i}{\sigma_a}$$

$$\theta_k = \frac{(2k+1)\,180^\circ}{n-m}, \quad k = 0, 1, \ldots, n-m-1$$

$$n - m = \frac{(2k+1)\,180^\circ}{\theta_k}$$

$$k = \frac{1}{2}\left(\frac{\theta_k\,(n-m)}{180^\circ} - 1\right)$$

$$n - m = 1 : \theta = 180^\circ$$

$$n - m = 2 : \pm 90^\circ$$

$$n - m = 3 : \pm 60^\circ,\ 180^\circ$$

PASO 6 · RUPTURA Y LLEGADA

$$K(\sigma) = -\frac{D(\sigma)}{N(\sigma)}$$

$$\frac{dK}{d\sigma} = -\frac{ND' - N'D}{N^2} = 0$$

$$\Rightarrow\ ND' - N'D = 0$$

$$N = 1 :$$

$$D'(\sigma) = 0$$

$$D = s^3 + a_2 s^2 + a_1 s + a_0$$

$$\Rightarrow\ D' = 3s^2 + 2a_2 s + a_1$$

$$\sigma_b = \frac{-a_2 \pm \sqrt{a_2^2 - 3a_1}}{3}$$

$$D = s^2 + a_1 s + a_0$$

$$\Rightarrow\ \sigma_b = -\frac{a_1}{2}$$

$$N = s + z :$$

$$(s+z)D'(s) - D(s) = 0$$

$$K_b = \left|\frac{D(\sigma_b)}{N(\sigma_b)}\right| = \frac{\prod |\sigma_b - p_i|}{\prod |\sigma_b - z_i|}$$

$$|D(\sigma_b)| = K_b\,|N(\sigma_b)|$$

$$\frac{d^2K}{d\sigma^2} < 0\ (\text{máximo}) \Rightarrow \text{ruptura}$$

$$\frac{d^2K}{d\sigma^2} > 0\ (\text{mínimo}) \Rightarrow \text{llegada}$$

PASO 7 · CRUCE CON JΩ (ROUTH)

$$s^3 + a_2 s^2 + a_1 s + a_0 = 0,$$

$$a_i = d_i + n_i K$$

$$\begin{array}{c|cc} s^3 & 1 & a_1 \\ s^2 & a_2 & a_0 \\ s^1 & b_1 & \\ s^0 & a_0 & \end{array}$$

$$b_1 = \frac{a_2 a_1 - a_0}{a_2}$$

$$b_1 = 0$$

$$\Rightarrow \; a_2 a_1 = a_0$$

$$N = 1:$$

$$K_{crít} = d_2 d_1 - d_0$$

$$N = s + z:$$

$$K_{crít} = \frac{d_0 - d_2 d_1}{d_2 - z}$$

$$a_2 s^2 + a_0 = 0$$

$$\Rightarrow \; \omega = \sqrt{\frac{a_0}{a_2}}$$

$$a_0 = a_2\,\omega^2$$

$$a_2 = \frac{a_0}{\omega^2}$$

$$\omega = \sqrt{a_1}$$

$$(\text{con } K = K_{crít})$$

$$a_1 = \omega^2$$

$$s = j\omega:$$

$$\text{Re}: a_0 - a_2\,\omega^2 = 0$$

$$\text{Im}: a_1\,\omega - \omega^3 = 0$$

$$0 < K < K_{crít} \text{ (estable)}$$

$$s^2 + a_1 s + a_0:$$

$$a_1 > 0,\; a_0 > 0$$

$$\Rightarrow \text{ estable } \forall K$$

PASO 8 · ÁNGULOS DE SALIDA Y LLEGADA

$$\theta_{sal} = 180^\circ + \sum \angle z_i - \sum \angle p_{\text{otros}}$$

$$\sum \angle p_{\text{otros}} = 180^\circ + \sum \angle z_i - \theta_{sal}$$

$$\sum \angle z_i = \theta_{sal} - 180^\circ + \sum \angle p_{\text{otros}}$$

$$\theta_{lleg} = 180^\circ + \sum \angle p_i - \sum \angle z_{\text{otros}}$$

$$\sum \angle z_{\text{otros}} = 180^\circ + \sum \angle p_i - \theta_{lleg}$$

$$\sum \angle p_i = \theta_{lleg} - 180^\circ + \sum \angle z_{\text{otros}}$$

$$\angle(q - p) = \tan^{-1} \frac{\text{Im}\, q - \text{Im}\, p}{\text{Re}\, q - \text{Re}\, p}$$

$$(+180^\circ \text{ si el denominador} < 0)$$

$$\angle(q - q^*) = 90^\circ$$

$$\theta_{sal}(q^*) = -\theta_{sal}(q)$$

PASO 9 · GRAFICACIÓN Y CALIBRACIÓN

$$\zeta = \frac{-\ln(\%OS/100)}{\sqrt{\pi^2 + \ln^2(\%OS/100)}}$$

$$\%OS = 100\, e^{-\zeta\pi/\sqrt{1-\zeta^2}}$$

$$\theta = \cos^{-1}\zeta$$

$$\zeta = \cos\theta$$

$$\tan\theta = \frac{\omega_d}{\sigma} = \frac{\sqrt{1-\zeta^2}}{\zeta}$$

$$\omega_d = \sigma \tan\theta$$

$$\sigma = \frac{\omega_d}{\tan\theta}$$

$$\omega_n = \sqrt{\sigma^2 + \omega_d^2}$$

$$\sigma = \zeta\,\omega_n$$

$$\omega_d = \omega_n\sqrt{1-\zeta^2}$$

$$\zeta = \frac{\sigma}{\omega_n}$$

$$\omega_n = \frac{\sigma}{\zeta}$$

$$s_0 = -\sigma + j\omega_d$$

$$\angle G(s_0) = \sum \angle(s_0 - z_i) - \sum \angle(s_0 - p_i) = -180^\circ$$

$$(\pm 360^\circ k)$$

$$\angle(s_0 + a) = \tan^{-1}\frac{\omega_d}{a - \sigma}$$

$$(+180^\circ \text{ si } a < \sigma)$$

$$|s_0 + a| = \sqrt{(a-\sigma)^2 + \omega_d^2}$$

$$K = \frac{\prod |s_0 - p_i|}{\prod |s_0 - z_i|} = \frac{1}{|G_1(s_0)|}, \quad G = K\,G_1$$

$$\prod |s_0 - p_i| = K \prod |s_0 - z_i|$$

$$\prod |s_0 - z_i| = \frac{\prod |s_0 - p_i|}{K}$$

$$T_s = \frac{4}{\sigma}$$

$$\sigma = \frac{4}{T_s}$$

$$T_p = \frac{\pi}{\omega_d}$$

$$\omega_d = \frac{\pi}{T_p}$$

$$D(s) + K\,N(s) = 0$$

$$\Rightarrow \text{ polos de lazo cerrado}$$